Structures Coursework

Group 8

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1 Span calculation

The beam is assessed by the virtual-work method. Span BC is solved first so that the support hinge moment at B and C can then be carried into spans AB and CD. The assumed plastic hinge sets are A-midspan-B for span AB, B-load point-C for span BC, and C-interior hinge for span CD.

1.1 Plastic analysis of span BC

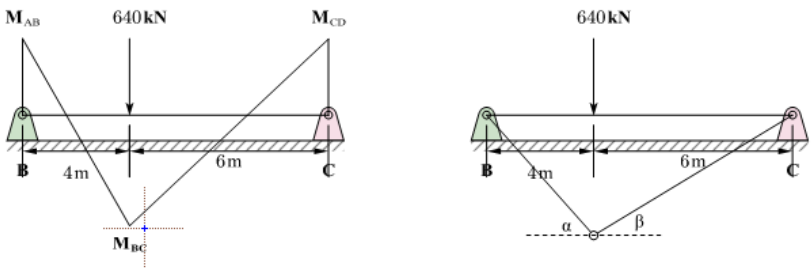


Figure 1: Collapse mechanism adopted for span BC.

Take plastic hinges at B, at the 640 kN load point, and at C. Let the required plastic moment for this span be $M_{p,BC}$. The span length is $BC = 10\,\text{m}$, with the load acting 4 m from B and 6 m from C.

Compatibility gives

$$4\theta_B = 6\theta_C \qquad \Rightarrow \qquad \theta_C = \frac{2}{3}\theta_B \qquad \text{and} \qquad \delta = 4\theta_B$$

The external work is

$$W_e = 640(4\theta_B) = 2560\theta_B$$

The internal work is

$$\begin{aligned} W_i &= M_{p,BC}\theta_B + M_{p,BC}\left(\theta_B + \frac{2}{3}\theta_B\right) + M_{p,BC}\left(\frac{2}{3}\theta_B\right) \\ &= \frac{10}{3}M_{p,BC}\theta_B \end{aligned}$$

At collapse,

$$2560\theta_B = \frac{10}{3}M_{p,BC}\theta_B$$

so

$$M_{p,BC} = \frac{2560 \times 3}{10} \qquad \Rightarrow \qquad M_{p,BC} = 768\,\text{kNm}$$

1.2 Plastic analysis of span AB

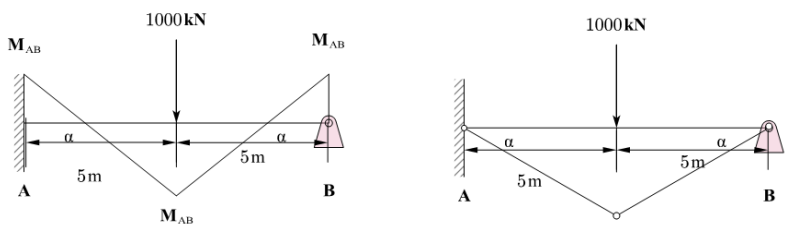


Figure 2: Collapse mechanism adopted for span AB.

For span $AB = 10\,\text{m}$, take hinges at A, at the midspan of AB, and at B. The support hinge at B is already fixed by the span BC mechanism, so $M_B = M_{p,BC} = 768\,\text{kN m}$. The fixed-end hinge at A and the positive hinge in span AB therefore both require moment $M_{p,AB}$.

Let the end rotations be θ. The deflection at the central point load $P = 1000\,\text{kN}$ is

$$\delta = 5\theta$$

The internal and external work are

$$\begin{aligned} W_i &= M_{p,AB}\theta + 2M_{p,AB}\theta + M_{p,BC}\theta \\ &= (3M_{p,AB} + 768)\,\theta \\ W_e &= 1000(5\theta) = 5000\theta \end{aligned}$$

At collapse, $W_i = W_e$, so

$$5000 = 3M_{p,AB} + 768 \qquad \Rightarrow \qquad M_{p,AB} \approx 1411\,\text{kNm}$$

1.3 Plastic analysis for span CD

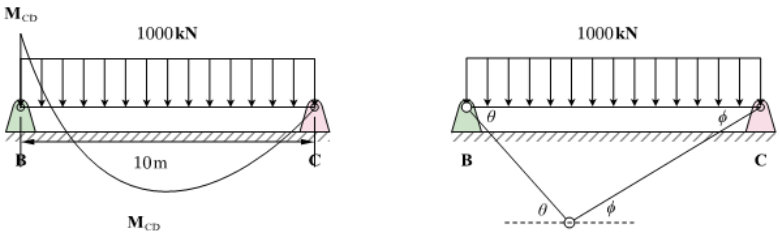


Figure 3: Collapse mechanism adopted for span CD.

For span CD, the negative support hinge at C is again taken from the span BC result, so $M_C = M_{p,BC} = 768\,\text{kN m}$. Let the positive hinge formed within the span require moment $M_{p,CD}$, and let its distance from C be x, while D remains the actual end support.

From compatibility,

$$x\theta = (12 - x)\phi \qquad \Rightarrow \qquad \phi = \frac{x}{12 - x}\theta$$

The external work under the uniform load $w = 100\,\text{kN/m}$ is

$$W_e = 100\left(\frac{1}{2} \cdot 12 \cdot x\theta\right) = 600x\theta$$

The internal work is

$$W_i = M_{p,BC}\theta + M_{p,CD}\,(\theta + \phi) = 768\theta + M_{p,CD}\theta\left(\frac{12}{12 - x}\right)$$

Equating external and internal work gives

$$600x = 768 + M_{p,CD}\left(\frac{12}{12 - x}\right)$$

so

$$M_{p,CD} = (600x - 768)\left(\frac{12 - x}{12}\right)$$

The critical mechanism is obtained from

$$\frac{dM_{p,CD}}{dx} = 0 \qquad \Rightarrow \qquad x = 6 + \frac{768}{1200} = 6.64\,\text{m}$$

Hence,

$$M_{p,CD} = (600(6.64) - 768)\left(\frac{12 - 6.64}{12}\right) \approx 1436\,\text{kNm}$$

Comparing the three spans, $M_{p,BC} = 768\,\text{kNm}$, $M_{p,AB} \approx 1411\,\text{kNm}$, and $M_{p,CD} \approx 1436\,\text{kNm}$, span CD governs, so $M_{p,\text{adopt}} \approx 1436\,\text{kNm}$.

1.4 Beam Sizing

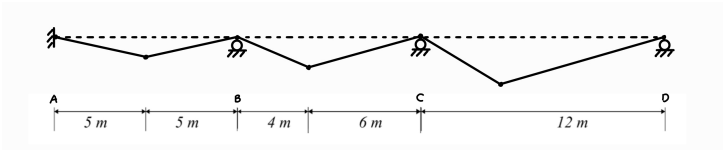


Figure 4: Beam Span

Assuming grade S355 steel with $f_y = 355\,\text{MPa}$, the beam is designed based on the governing plastic moment capacity from span CD, where $M_{p,\text{adopt}} \approx 1436\,\text{kNm}$, which is adopted consistently throughout the beam in Figure 4.

The required plastic section modulus is obtained from:

$$W_{pl,req} = \frac{M_p\,\gamma_{M0}}{f_y}$$

Assuming $\gamma_{M0} = 1.0$,

$$W_{pl,req} = \frac{1.436 \times 10^9 \times 1.0}{355}$$

$$W_{pl,req} \approx 4.05 \times 10^6\ \text{mm}^3$$

Referring to official British Steel section tables, a UB 610x229x140 section with $W_{pl} = 4.20 \times 10^6\ \text{mm}^3$ is adopted for the full beam [3]. This exceeds the required plastic modulus while remaining close to the originally considered 610-series beam depth, so it is still practical to fabricate, handle and connect.

A uniform section is selected rather than reducing the section in span BC and introducing transitions, as discrete section changes and associated connections can introduce local deviations from idealised beam behaviour and reduce the reliability of stiffness and plastic capacity distribution [4]. Maintaining a constant section ensures a more predictable global response and a clearly defined plastic collapse mechanism.

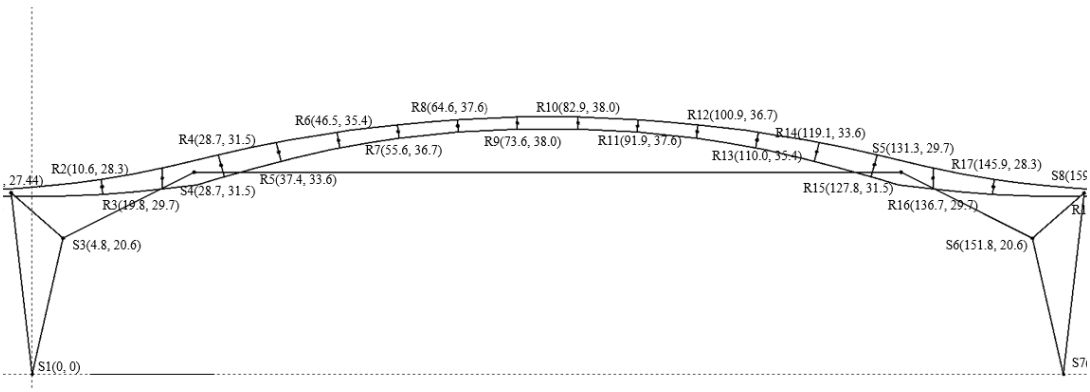
Furthermore, adopting a single rolled section removes the need for moment-resisting splices or transition detailing, simplifying fabrication and erection. Since fabrication and installation typically account for a significantly larger proportion of total project cost than material alone [1], this improves construction efficiency and offsets any marginal material over-design in span BC.

2 Deliverable 1: Geometry

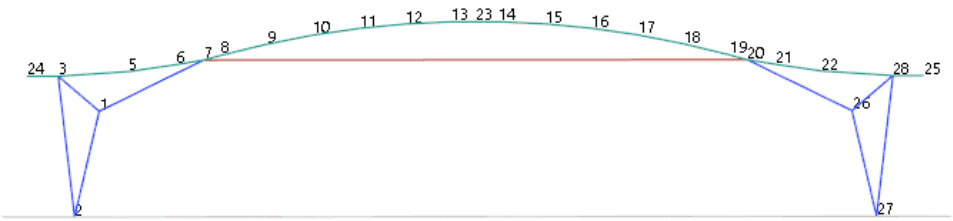
The roof geometry was reconstructed from the supplementary drawing using a one-to-one SketchUp model and then transferred into GSA. As shown in Figure 5(a), points labelled R define the roof profile and represent the mid-points of the roof section, while points labelled S define the support system and lower restraint points.

The second diagram shows the FE idealisation adopted for analysis. In Figure 5(b), green denotes beam elements used for the roof profile, red denotes the high tie modelled as a tie element, and blue denotes the support-tree members modelled as bar elements. In GSA, beam elements model axial, bending and torsional effects, bar elements provide axial stiffness only, and tie elements are axial-only elements with zero stiffness in compression. This makes beams suitable for the roof rafters, bars suitable for the tree members, and a tie suitable for the prestressed high tie. Both end supports were modelled as fixed supports, while the pin-connected members were treated consistently on both sides because the connected elements were assumed not to transfer bending moment. This idealisation captures the main load path and preserves the key dimensions required for later loading, validation and mesh sensitivity studies.

Because both the arch rafter and tusk rafter vary in depth along their length, the roof-profile members were implemented in GSA using tapered 1D beam elements rather than prismatic sections. Oasys GSA states that a 1D element can taper between end 1 and end 2 when it is assigned a tapering section profile with two section sizes, and that when a tapering member is divided into multiple elements a taper offset is used so that each element retains the correct local dimensions from its position along the original member [5]. This is important for the present roof model because the varying rafter depth changes the member bending stiffness along the span, so representing that taper improves the fidelity of the global frame idealisation.



(a) Reconstructed nodal geometry



(b) Idealised FE member types

Figure 5: Geometry adopted for the global model.

3 Deliverable 2: Material, Section, and Loading Data

3.1 Material and Section Properties and Loading Data

The global FE model was defined using linear-elastic steel. In accordance with the brief, all members were assigned an elastic modulus of 200 GPa, a Poisson's ratio of 0.30 and density of 7850 kg m^{-3} were adopted so that self-weight could be included automatically in GSA. S235 steel was adopted for the primary roof members because its elastic modulus complies with the brief requirement, while other material parameters such as shear modulus and strength are not governing in this linear-elastic global analysis. The twin high-tie strands were modelled as a single prestressed tie element with the same total cross-sectional area and an initial prestress of 450 MPa.

Only vertical symmetric loading was considered. The characteristic permanent roof load was taken as the sum of the aluminium standing seam roof (0.10 kPa), structural deck (0.15 kPa), insulation/acoustic layers (0.40 kPa) and the service allowance (0.15 kPa), giving $G_k = 0.80$ kPa. The live load imposed on the roof was taken as $Q_k = 0.60$ kPa. The governing vertical ultimate limit state (ULS) combination used in the model was

$$1.35G_k + 1.5Q_k \tag{1}$$

Here, G_k and Q_k were converted to line loads using the projected tributary width carried by one transverse frame. The supplementary roof information states that the main rafters and associated tie-rods are spaced at 18 m centres, so a tributary width of 18 m was adopted for the 2D frame idealisation. On this basis, the superimposed permanent roof load is 14.4 kN m^{-1} (0.80×18) and the imposed live load is 10.8 kN m^{-1} (0.60×18), with member self-weight still included separately through the material density. This projected-area treatment is consistent with common roof-loading practice and with the basis commonly used for roof snow loading in Eurocode 1 [2].

The strap members were modelled as bar elements. Therefore, only their axial stiffness was represented in the global model, using an equivalent steel area of $A = 2 \times 500 \times 40 = 40\,000$ mm^2. The spacing between the two plates was neglected in this idealisation because the bar elements do not carry bending.

The adopted member sections and the associated modelling notes are summarised in Table 1.

Table 1: Section data adopted for the global model

Member	Section	Modelling Note
Arch rafter	Welded box, 800 mm wide, 50 mm wall thickness	Depth varies from about 1.8 m to 3.8 m along the roof profile
Tusk rafter	Welded I-section, 50 mm plate thickness	Overall depth varies from about 0.9 m to 3.8 m
Legs and arms	Circular Hollow Section (CHS) 914 mm × 40 mm	Taken directly from the supplementary sketch
Wings	CHS 711 mm × 50 mm	Taken directly from the supplementary sketch
High tie	Twin locked-coil strands, 2 × 115 mm	Modelled as one equivalent tie element with the same total cross-sectional area; prestressed to 450 MPa
Straps	Equivalent rod area 40 000 mm^2	Based on two 500 mm × 40 mm plates; bending stiffness neglected

4 Deliverable 3: Initial Model Validation

4.1 Assumptions

The validation assumes a 2D in-plane frame subjected to vertical symmetric loading. Out-of-plane bending, torsion, adjacent-frame load sharing and detailed connection stiffness are neglected, so the check is intended only to confirm that the global model gives a realistic level of response rather than an exact local solution. This 3D-to-2D simplification is most reasonable near the middle of the roof, where out-of-plane bending moments are expected to remain limited under symmetric loading, but it may be less accurate near the roof edges where edge effects can make out-of-plane bending more significant. Torsion is expected to be a secondary effect because both the applied loading and the main structural arrangement are predominantly symmetric. Lateral buckling and shear buckling are also not represented in the present global model, so these stability modes would need to be checked separately in a more detailed assessment.

4.2 Model Validation

As the prestressed arch-and-tie behaviour is difficult to analyse by simple hand calculations, the initial validation was carried out on the non-prestressed model and focused on the bending moment at the tusk-to-arch connection. The loaded tusk was idealised as a 25 m cantilever under the equivalent ULS line load $w = 1.35 \times 14.4 + 1.5 \times 10.8 = 35.64\,\mathrm{kN\,m^{-1}}$, giving a tusk-to-arch connection moment $M = wL^2/2 \approx 11\,125\,\mathrm{kN\,m}$, as shown in Figure 6. The corresponding FE result is about 16 490 kN m, so the model predicts the correct order of magnitude for this key bending action. The difference is expected because the hand check neglects the stiffness of the curved arch, frame interaction between both sides of the roof, and the more complex redistribution captured by the FE model.

An additional source of modelling error was observed at the high-tie connection. The FE model predicts a relatively large vertical deformation there, even though the vertical nodal force associated with the high tie is small compared with the forces transferred by the surrounding members. Using the largest adjacent vertical force from the connection check, the high-tie vertical force is only about $114.6/5901 \approx 1.9\%$ of the neighbouring member force. This suggests that the large vertical movement is more likely to arise from compatibility within the simplified 2D global model and the axial-only idealisation of the tie than from a large vertical load carried by the high tie itself.

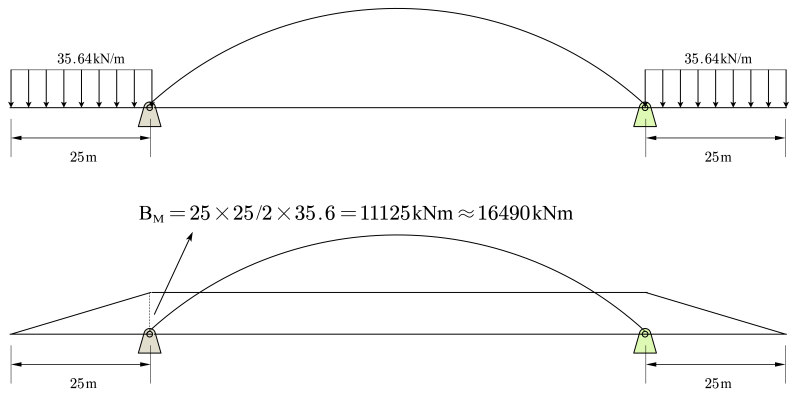


Figure 6: Hand validation of the tusk-to-arch connection bending moment.

4.3 Model Adequacy

Basic verification checks were also completed: the GSA solver ran without instability warnings, the vertical reactions balanced the applied gravity loads, and the reconstructed model matched the original drawing on both sides with an overall geometric deviation of approximately 0.5%. Taken together, the modelling assumptions, validation considerations and verification checks indicate that the global frame model is adequate for an initial study of the roof under symmetric vertical loading. However, it should still be understood as a simplified global model rather than a full representation of the three-dimensional structural behaviour.

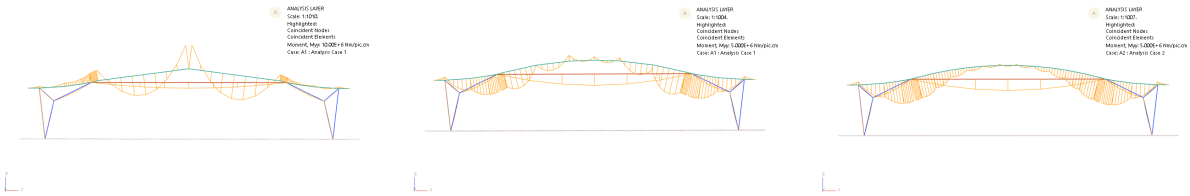
5 Deliverable 4: Mesh Refinement Sensitivity Study

5.1 Convergence Study Methodology

Three mesh densities were evaluated under the governing ULS vertical load combination $1.35G_k + 1.5Q_k + 1.0P_s$, where P_s represents the prestress in the high-tie. Mesh 1 is the coarsest representation of the arch, Mesh 2 retains more intermediate points, and Mesh 3 preserves the full coordinate set taken from the supplementary geometry. Arch central deflection, high-tie axial force and arch central bending moment were tracked to judge convergence.

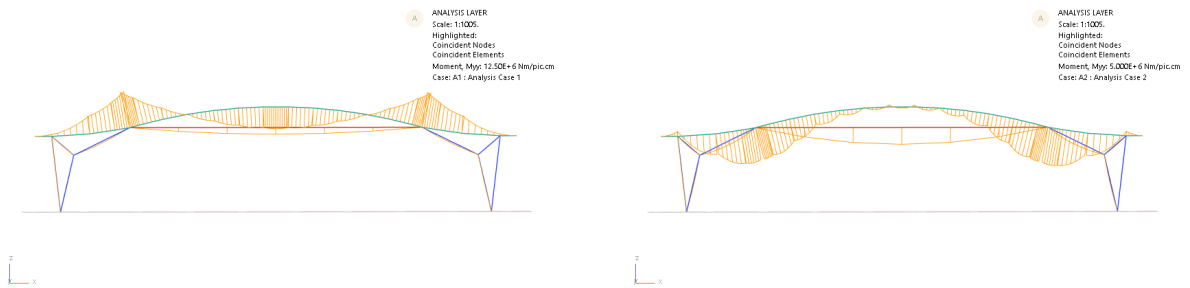
Table 2: Influence of arch mesh density on key global responses

Mesh	Arch Elements	DoF	Arch Central Deflection (mm)	High-Tie Axial Force (kN)	Arch Central Bending Moment (kNm)	% Deflection Change vs Mesh 3
Mesh 1	14	53	241.2	9725	12760	37.6%
Mesh 2	16	59	175.7	9309	960.2	0.2%
Mesh 3	22	77	175.3	9320	198.2	0.0%



Mesh 1 Mesh 2 Mesh 3

Figure 7: Bending-moment diagrams for the three mesh densities used in the sensitivity study.



Prestress neglected Prestress included

Figure 8: Comparison of bending-moment response with and without high-tie prestress.

Table 2 and Figures 7 and 8 show that the coarsest arch discretisation is too flexible. Relative to Mesh 3, Mesh 1 gives much larger deflection and a very different bending-moment pattern. As the arch is reduced towards only a few straight beam segments, its curvature and therefore its stiffness are lost, so the response moves closer to the non-prestressed case: the frame behaves less like a stiff prestressed arch and more like a softer flexural system. Mesh 2 is already close to Mesh 3 for deflection and tie force, but the central bending moment remains much more sensitive to curvature, so Mesh 3 is adopted for the remainder of the report.

5.2 Conclusion

Mesh 3 is adopted for the remainder of the report.

6 Deliverable 5: Influence of Prestressing Assumptions

6.1 Quantitative Comparison

Table 3: Simplified sensitivity of global response to high-tie prestress

Response Quantity	**Prestress Neglected**	**Prestress Included**	**Change With Prestress**
Maximum vertical displacement (mm)	490	175.3	−64.2%
Centre rafter bending moment (kNm)	9811	198.2	−98.0%
Tusk-to-arch connection bending moment (kNm)	16490	5837	−64.6%
High tie force (kN)	4296	9320	+116.9%
Horizontal reaction at supports (kN)	4211	3480	−17.4%

6.2 Discussion of Results

Table 3 and Figure 8 show that prestress changes the global response markedly. Without prestress, the central arch sags under gravity and the two tusks rotate upward to maintain continuity at the arch-tusk junctions. With prestress, the central arch retains a clearer curved profile, so the deformation pattern reverses and the tusks are pushed downward. When the arch sags more, the roof behaves in a more flexural manner and the moment distribution can change sign over parts of the frame, whereas the prestressed case remains closer to arch action.

6.3 Implications for Design and Tolerance

These results show that prestress is critical to design interpretation and tolerance assessment. If prestress is neglected, the model overestimates deflection and rafter bending but underestimates the tensile demand in the high tie and its anchorage. This is consistent with the supplementary construction sequence, where staged stressing was used to control geometry and force state, so the prestressed model is adopted as the minimum realistic representation of the completed roof. The large change in response also indicates that, without prestress, the arch would be much less stiff and more difficult to keep within a realistic construction-tolerance envelope.

7 Deliverable 6: Boundary Conditions for More Detailed Analysis

For a more detailed FE study, the rafter splice between the tusk rafter and arch rafter should be extracted from the global frame model and analysed as a local submodel. The purpose of the local model would be to resolve the stress distribution in the splice plates, bolts or welds, and the adjacent box and I-section walls, which cannot be captured accurately by the beam idealisation used in the global model.

The local model should include a short length of member on each side of the splice so that the boundary conditions are applied away from the connection itself. This reduces artificial stress concentrations caused by restraining the model too close to the detail. A practical choice would be to include approximately one section depth of the arch rafter and one section depth of the tusk rafter beyond the splice on either side.

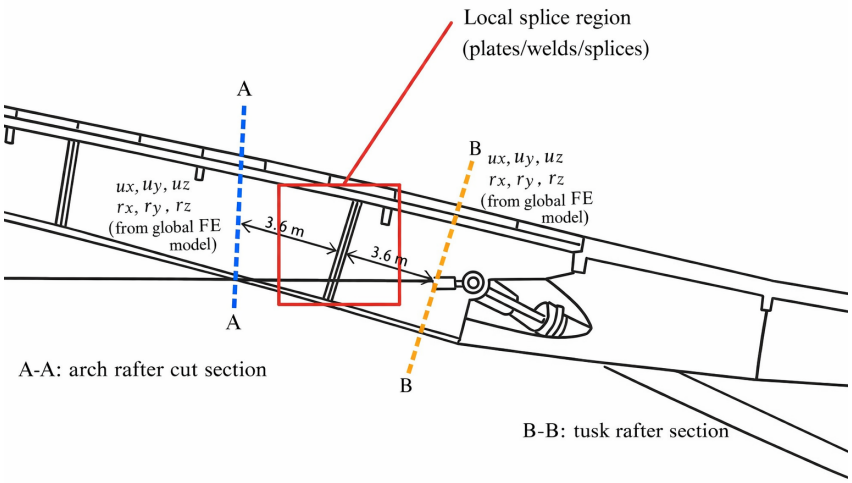


Figure 9: Recommended boundary conditions for the local splice model

The preferred boundary condition is to impose the displacement and rotation components extracted from the global beam model at the two cut faces of the local model. In practice, each cut face can be coupled to a reference node so that the sectional deformation remains compatible with the global model. The six kinematic quantities imposed at each end are the three translations (u_x, u_y, u_z) and the three rotations (r_x, r_y, r_z). This approach is preferable to simply fixing one end, because it reproduces the actual deformation state of the connection and avoids introducing artificial restraint moments.

The corresponding section forces should also be checked against the global model outputs. The main actions expected at the splice are axial force N, vertical shear V, bending moment M, and a smaller torsional component due to the curved roof geometry and eccentricity of the connection detail. If the software or modelling approach makes direct displacement transfer difficult, an acceptable alternative is to apply the equivalent set of sectional resultants at the two cut faces, provided that overall equilibrium is maintained.

To prevent rigid body motion in the local model, one reference node may be lightly stabilised after the global displacements have been assigned. This stabilisation should only remove numerical singularities and must not alter the transferred load path. Contact, bolt preload, or weld representation can then be added depending on the intended level of detail. For the purposes of this coursework, the key requirement is that the splice boundaries are driven by the global model response rather than by arbitrary fixed supports.

Section cuts are located approximately one local section depth away from the splice connection along the axis of the member. This avoids local stress disturbance. As the depth varies along the rafter, the local section depth is equal to the splice-region depth (3.6 m), cuts A–A and B–B are placed approximately 3.6 m from the splice on either side. At both cut faces, the translational and rotational displacements extracted from the global FE model are imposed. This ensures compatibility between the local splice model and the global structural response.

References

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